

CRLA Multi-Year Reading Proficiency Analysis

1. Abstract

The Classroom Reading Level Assessment (CRLA) is the Department of Education's standardized tool for measuring early-grade reading proficiency across Philippine public schools. This project implements a reproducible analytical pipeline that harmonizes raw CRLA data across school years, quantifies each school's proficiency distribution through three statistical moments (mean, standard deviation, and skewness), tracks progress between consecutive assessment periods, and produces a principled priority ranking of schools for intervention targeting.

The pipeline currently covers three timepoints — BoSY 2024-25, EoSY 2024-25, and BoSY 2025-26 — spanning approximately 39,000 schools and over 10 million assessed learners in Grades 1-3. National results show a within-year gain of +0.82 ordinal levels (BoSY to EoSY 2024-25) followed by a cross-year decline of -1.04 (EoSY 2024-25 to BoSY 2025-26), yielding a net regression of -0.14 across the observation window.

To answer where interventions should be focused, the pipeline combines each school's academic need, student population size, and local government fiscal capacity into a composite priority score. Sensitivity analysis confirms that the resulting priority list is robust to methodological choices: the top-100 intervention targets are stable across 500 random weight configurations, with 60% of all ranked schools showing near-zero rank volatility.

2. Background

CRLA classifies each assessed learner into one of five reading proficiency levels, ordered from lowest to highest:

Level	Abbreviation	Ordinal Value
Lower Emergent	LE	1
Higher Emergent	HE	2
Developing	Dev	3
Transitioning	Trans	4
Grade Level	GL	5

Assessments are administered at the beginning (BoSY) and end (EoSY) of each school year, across three grade levels and six grade-language groups:

Grade-Language Group	Description
G1	Grade 1 (single assessment)
G2 MT	Grade 2, Mother Tongue
G2 Fil	Grade 2, Filipino
G3 MT	Grade 3, Mother Tongue
G3 Fil	Grade 3, Filipino
G3 Eng	Grade 3, English

Each school's raw data consists of student counts per proficiency level for each grade-language group, yielding 30 columns per timepoint.

The central policy questions this analysis addresses:

1. Are learners improving across assessment periods?
2. V

3. Data

3.1 CRLA Assessment Files

Timepoint	File	Schools
BoSY 2024-25	CRLA Results Archive_SY 2024-25	35,280
	Assessment Results_Table_BoSY.csv	
EoSY 2024-25	CRLA Results Archive_SY 2024-25	37,045
	Assessment Results_Table_EoSY.csv	
BoSY 2025-26	CRLA National Dashboard_BoSY	38,965
	2025-26 Assessment Results_Table.csv	

3.2 External Data Sources

Source	File	Purpose
DepEd School Level Database with PSGC	SY 2024-2025 School Level Database WITH PSGC.xlsx	Maps each School ID to a Philippine Standard Geographic Code (PSGC) municipality/city code
DOF BLGF Statement of Receipts and Expenditures	By-LGU-SRE-2024.xlsx	
Public enrollment (Project Bukas)	public_project_bukas_enrollment_2024-25.csv	School-level enrollment counts used to compute SEF per enrolled learner at the LGU level

All input files are located in data/raw/.

3.3 Schema Hazards

The raw CRLA CSVs from different school years exhibit several inconsistencies that the pipeline resolves automatically:

1. Column order differences: 2024-25 places G2 Fil Transitioning before G2 Fil Developing; 2025-26

Reference: [documentation/step_1_schema_harmonizer.md](#)

4. Methodology

4.1 Schema Harmonization

Raw CSVs from different school years are normalized into an identical column schema. Column matching is name-based, not positional, which avoids column-swap bugs present in earlier positional (iloc) approaches. Operations include header typo correction, canonical column ordering of the 30 reading profile columns, removal of aggregate Total columns, and mojibake repair.

Reference: [documentation/step_1_schema_harmonizer.md](#)

4.2 Ordinal Proficiency Scoring

For each school at each timepoint, raw student counts are first converted to percentages within each grade-language group, then summarized through three statistical moments that together characterize the school's proficiency distribution.

4.2.1 Percentage Conversion

Within each of the six grade-language groups, raw counts are divided by the group total to obtain the proportion of learners at each proficiency level. Groups with zero assessed learners yield missing values.

4.2.2 Ordinal Mean

The ordinal mean is a weighted average of the proficiency distribution, averaged across all grade-language groups with data:

$$\bar{x}_s = \frac{1}{G} \sum_{g=1}^G \sum_{k=1}^5 p_{s,g,k} \cdot w_k$$

where p is the percentage of learners in school s , grade-language group g , at proficiency level k , and $w \in \{1, 2, 3, 4, 5\}$ are the ordinal values from Lower Emergent to Grade Level. The result is a score on the 1-5 scale: a value of 3.30 means the average learner in that school falls between the Developing and Transitioning levels.

Why ordinal scoring over PCA. An alternative PCA-based scoring method was explored, where weights are derived from the first principal component of the proficiency distribution. PCA weights proved unstable across reference timepoints — Grade Level receives weight 0 under a BoSY reference but weight 100 under an EoSY reference. The ordinal method uses fixed, interpretable weights that require no reference period.

Property	PCA Scoring	Ordinal Scoring
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Scale	0-100 (arbitrary)	
Reference timepoint	Required	Not needed
Stability across timepoints	Weights change with reference	Fixed
Interpretability	Opaque	"Average proficiency level"

4.2.3 Ordinal Standard Deviation

The ordinal standard deviation measures how spread out a school's learners are across proficiency levels:

$$\sigma_s = \sqrt{\frac{1}{G} \sum_{g=1}^G \sum_{k=1}^5 p_{s,g,k} \cdot (w_k - \bar{x}_{s,g})^2}$$

A low SD indicates learners are clustered around the mean level; a high SD indicates a wide spread (e.g., many learners at both Grade Level and Lower Emergent). Higher SD signals greater within-school inequality.

4.2.4 Ordinal Skewness

The ordinal skewness captures asymmetry in the proficiency distribution:

$$\gamma_s = \frac{1}{G} \sum_{g=1}^G \frac{\sum_{k=1}^5 p_{s,g,k} \cdot (w_k - \bar{x}_{s,g})^3}{\sigma_s^3}$$

Negative skewness (the national tendency) indicates a long lower tail — most learners are at higher levels but a subset remains at lower ones. Positive skewness indicates the reverse. Schools with zero SD (all learners at one level) have undefined skewness, treated as missing.

Reference: [documentation/step_2_percentages_and_scoring.md](#)

4.3 Validation

Each school is validated at two tiers. The basic tier requires at least one grade-language group with non-missing data. The strict tier, used for priority ranking, imposes three additional requirements:

Criterion	Threshold	Rationale
Grade coverage	All three grade levels (G1, G2, G3) have at least one reporting group	Ensures the school's score reflects the full early-grade span
Group breadth	At least 4 of 6 grade-language groups reporting	Guards against scores driven by a single group
Minimum sample	At least 15 assessed learners in every reporting group	Ensures statistical stability of within-group percentages

A segment delta between two timepoints additionally requires: (1) both endpoints pass basic validation, and (2) the total assessed student count does not change by more than 25% between endpoints.

$$\text{valid_segment} = \text{valid}_{t_0} \wedge \text{valid}_{t_1} \wedge \left| \frac{N_{t_1} - N_{t_0}}{N_{t_0}} \right| \leq 0.25$$

The count-stability check guards against school mergers, splits, or data-entry errors that would make a progress comparison misleading.

4.4 Chain-Based Progress Scoring

An ordered time chain defines the sequence of assessment periods:

$$t_1 \rightarrow t_2 \rightarrow t_3 \quad = \quad \text{BoSY 2024-25} \rightarrow \text{EoS Y 2024-25}$$

For each consecutive pair in the chain, a segment delta is the change in ordinal mean:

$$\Delta_i = \bar{x}_{t_{i+1}} - \bar{x}_{t_i}$$

The composite progress score is the sum of all valid segment deltas, representing cumulative progress across the full chain:

$$\Delta_{\text{composite}} = \sum_i \Delta_i$$

Segment deltas are also computed for SD and skewness, tracking how distributional shape changes over time.

The pipeline extends automatically when new timepoints are added — appending a new entry to the time chain produces new segment deltas without modifying any existing computations.

Reference: [documentation/step_3_chain_progress.md](#)

4.5 LGU Matching

To incorporate local government fiscal data into the priority ranking, each school must be linked to its municipality or city. The pipeline builds a school-to-LGU crosswalk by joining three data sources:

1. CRLA data provides the School ID.
2. D

Coverage: 1,631 of 1,634 cities/municipalities matched (99.8%), covering 99.0% of CRLA schools.

Reference: [documentation/lgu_matching.md](#)

4.6 Priority Ranking

Schools passing strict validation are ranked for intervention targeting using a three-pillar composite. Each pillar captures a distinct dimension of priority:

Pillar 1 — Need. A weighted sum of standardized (z-scored) ordinal moment components:

$$\text{Need}_s = \sum_j \alpha_j \cdot z(c_{s,j})$$

where c are six need components and α are their weights:

Component	Weight	Direction	Interpretation
5 – mean	2.0	Lower mean = higher need	Current proficiency level

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–delta mean	2.0	Declining trajectory = higher need	Direction of change
SD (endpoint)	1.0	Higher spread = higher need	Within-school inequality
Skewness (endpoint)	1.0	Positive skew = higher need	Distribution shape
delta SD	0.5	Increasing spread = higher need	Change in inequality
delta skewness	0.5	Worsening skew = higher need	Change in distribution shape

Pillar 2 — Impact. The total number of assessed learners at the segment endpoint. Schools with more students represent a larger potential impact per intervention.

Pillar 3 — Capacity Gap. The inverse of the LGU's Special Education Fund (SEF) per enrolled learner:

$$\text{SEF per capita} = \frac{\text{LGU Special Education Fund}}{\text{Total enrolled learners in LGU}}$$

The SEF is a component of Real Property Tax revenue earmarked for education. Lower SEF per capita indicates the LGU has fewer resources available per student, representing a greater capacity gap. Enrollment counts are sourced from the SY 2024-25 public enrollment dataset, aggregated to the LGU level via the school-PSGC crosswalk.

Composite. Each pillar is converted to a percentile rank, then the composite is their product:

$$\text{Priority}_s = R_{\%}(\text{Need}_s) \times R_{\%}(\text{Impact}_s) \times R_{\%}(\text{CapacityG}_s)$$

Percentile ranks map each pillar to a uniform [0, 1] distribution regardless of its raw distribution shape. This is critical because assessed counts (skewness ≈ 5) and SEF per capita are heavily right-skewed — without this transformation, a handful of outlier schools would dominate the composite. The multiplicative form ensures a school must score high on all three dimensions to rank at the top.

Reference: [documentation/priority_ranking.md](#)

4.7 Sensitivity Analysis

The Need pillar weights α are chosen by domain judgment, not estimated from data. To assess how much the final rankings depend on these choices, two analyses are conducted:

Scenario comparison. Five interpretable weight profiles — default, level-focused, delta-focused, mean-only, and equal — are compared using Kendall τ rank correlation (across all ranked schools) and Jaccard overlap (of the top-N lists).

Per-school robustness. 500 random weight profiles are drawn from a Dirichlet distribution ($\alpha = 2.0$). For each school, the analysis reports the median priority percentile, interquartile range (IQR) of priority percentile, and the fraction of draws in which the school appears in the top-100. Schools with $\text{frac_top_100} = 100\%$ are robust intervention targets whose ranking does not depend on the weight specification.

Reference: [documentation/sensitivity_analysis.md](#)

5. Key Results

5.1 National Proficiency Trends

Across 29,854 schools with valid data at all three timepoints:

Timepoint	Mean	SD	Skewness	Interpretation
BoSY 2024-25	3.30	1.07	-0.49	Between Developing and Transitioning
EoSY 2024-25	4.13	0.87	-1.18	
BoSY 2025-26	3.06	1.08	-0.31	Just above Developing

Segment	Mean Delta	Direction
BoSY 2024-25 → EoSY 2024-25	+0.82	Within-year improvement
EoSY 2024-25 → BoSY 2025-26	-1.04	Cross-year decline
Composite	-0.14	Slight net regression

The within-year gain indicates that schools moved learners upward by nearly a full proficiency level on average during SY 2024-25. However, the cross-year decline (summer learning loss plus new-cohort effects) more than offset this gain, resulting in a small net regression over the full observation window.

5.2 Validation Coverage

Tier	BoSY 2024-25	EoSY 2024-25	BoSY 2025-26
Basic (at least 1 group with data)	35,280 (100%)	37,042 (>99%)	38,964 (>99%)
Strict (all grades, ≥4 groups, ≥15 per group)	22,961 (65%)	24,380 (66%)	24,303 (62%)
Full chain, strict (all 3 timepoints)	19,152 (49%)		

5.3 Priority Ranking

Segment	Strict-Valid	Ranked	Dropped (missing LGU data)
BoSY → EoSY 2024-25	22,206	22,114	92
EoSY 2024-25 → BoSY 2025-26	20,855	20,766	89

Capacity gap statistics (SEF per capita): national median = PHP 355; mean = PHP 954.

5.4 Sensitivity Analysis

Scenario comparison (Segment 1):

Pair	Kendall τ	Top-100 Jaccard
Default vs Mean-only	0.917	88.7%
Default vs Equal	0.935	83.5%
Default vs Level-focused	0.858	66.7%
Default vs Delta-focused	0.820	57.5%

Robustness (500 Dirichlet draws, Segment 1):

Stability Tier	IQR Threshold	Schools	Share
Stable	$IQR < 0.05$	10,703	60%
Moderate	$0.05 \leq IQR < 0.15$	6,517	37%
Volatile	$IQR \geq 0.15$	616	3%

The overall rank ordering is robust ($\text{Kendall } \tau > 0.82$ across all scenario pairs). The main axis of sensitivity is the relative emphasis on current proficiency levels versus trajectory — not the inclusion of SD or skewness. Schools appearing in the top-100 across 100% of random weight draws are the strongest intervention candidates.

5.5 Validation Against Prior School List

A previously curated list of 131 priority schools (identified using the earlier PCA-based method) was compared against the current pipeline's Learning segment (BoSY → EoSY 2024-25) priority ranking. All 131 School IDs are unique (authoritative source: `output/bbi_annex_b_school_ids.txt`). Of these, 66 passed strict validation and appeared in the current ranking, while 65 were excluded due to insufficient data — primarily too few assessed learners for reliable ordinal statistics.

Reference: `documentation/old_list_comparison.md`

6. Repository Structure

<code>project_crla/</code>	
<code> </code>	
<code> — modules/</code>	Analysis pipeline
<code> — preprocessing.py</code>	Schema harmonization, data loading, percentage conversion, validation
<code> — analysis.py</code>	Ordinal moments, performance scoring, chain-based progress
<code> — output.py</code>	Pairwise CSV export
<code> — priority_ranking.py</code>	Three-pillar priority ranking
<code> — sensitivity_analysis.py</code>	Need weight sensitivity and robustness
<code> — lgu_matching.py</code>	School-to-LGU crosswalk and revenue matching
<code> — gcs_utils.py</code>	Google Cloud Storage paths and filesystem
<code> — crla_v2.py</code>	Original reference implementation (legacy)
<code> — documentation/</code>	Detailed methodology references

		multi_year_expansion_plan.md	Overall pipeline design and rationale
		step_1_schema_harmonizer.md	Schema normalization details
		step_2_percentages_and_scoring.md	Scoring methodology comparison
		step_3_chain_progress.md	Chain-based progress scoring
		step_4_output.md	Output format and verification
		priority_ranking.md	Three-pillar ranking methodology
		sensitivity_analysis.md	Weight sensitivity and robustness results
		lgu_matching.md	School-to-LGU crosswalk methodology
		old_list_comparison.md	131-school legacy list validation
		policy_brief_crla_methodology.md	One-page policy brief
		dashboard_plan.md	Streamlit dashboard specifications
		notebooks/	Stakeholder presentation notebooks
			1.0_methodology_walkthrough.ipynb Three-pillar methodology explainer
			2.0_portfolio_analysis_131_schools.ipynb Legacy list re-evaluation
			3.0_cycle2_interpretability.ipynb Cycle 2 interpretability analysis
		dashboard/	Interactive Streamlit dashboard
			app.py Application router
			pages/ Dashboard views
			data/ Pre-computed dashboard data
		data/	
			raw/ Input CSVs and external data files
			modified/ Crosswalks and reference files
		output/	Pipeline outputs
			priority_ranking_sef.*.csv Priority rankings by segment
			sensitivity_*.csv Sensitivity analysis outputs
			crla_progress_score.*.csv Pairwise progress CSVs
		README.md	

7. Reproducibility

7.1 Dependencies

- Python 3.11+

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7.2 Running the Pipeline

1. Load and harmonize data:

```
import sys
sys.path.insert(0, 'modules')
from preprocessing import load_all_assessments

file_map = {
    ('2024-25', 'BoSY'): 'data/raw/CRLA Results Archive_SY 2024-25 Assessment Results_Table_BoSY.csv',
```

```
( '2024-25', 'EoS' ): 'data/raw/CRLA Results Archive_SY 2024-25 Assessment Results_Table_EoS.csv',
( '2025-26', 'BoS' ): 'data/raw/CRLA National Dashboard_BoS 2025-26 Assessment Results_Table.csv',
}
df_all = load_all_assessments(file_map=file_map, source='local')
```

2. Compute ordinal moments and chain progress:

```
from analysis import process_all_timepoints, compute_chain_progress, TIME_CHAIN

results = process_all_timepoints(df_all, scoring='ordinal')
progress_df = compute_chain_progress(
    performance=results['performance'],
    raw_data=df_all,
    validation=results['validation'],
    time_chain=TIME_CHAIN,
    ordinal_sd=results['ordinal_sd'],
    ordinal_skew=results['ordinal_skew'],
)
```

3. Export pairwise progress CSVs:

```
from output import export_pairwise_csvs

export_pairwise_csvs(
    percentages=results['percentages'],
    performance=results['performance'],
    validation=results['validation'],
    raw_data=df_all,
    weights=results['weights'],
    time_chain=TIME_CHAIN,
    output_dir='output',
)
```

4. Generate priority rankings:

```
from lgu_matching import load_deped_psgc, load_lgu_revenue, build_school_lgu_crosswalk, match_lgu_revenue
from priority_ranking import compute_priority_ranking

psgc_df = load_deped_psgc('data/raw/SY 2024-2025 School Level Database WITH PSGC.xlsx')
crosswalk_df = build_school_lgu_crosswalk(psgc_df)
lgu_revenue_df = load_lgu_revenue('data/raw/By-LGU-SRE-2024.xlsx')
matched_lgu_df, unmatched_df, match_log = match_lgu_revenue(crosswalk_df, lgu_revenue_df)

for seg_idx in range(len(TIME_CHAIN) - 1):
    ranking_df, summary = compute_priority_ranking(
        progress_df, seg_idx, TIME_CHAIN, crosswalk_df, matched_lgu_df,
    )
    ranking_df.to_csv(f'output/priority_ranking_seg.{summary["segment"]}.csv')
```

5. Run sensitivity analysis:

```
from sensitivity_analysis import run_scenario_comparison, run_robustness_analysis

for seg_idx in range(len(TIME_CHAIN) - 1):
    scen = run_scenario_comparison(progress_df, seg_idx, TIME_CHAIN, crosswalk_df, matched_lgu_df)
    rob = run_robustness_analysis(progress_df, seg_idx, TIME_CHAIN, crosswalk_df, matched_lgu_df)
```

7.3 Data Access

Raw input files can be loaded from local paths (as shown above) or from Google Cloud Storage. The default configuration in `modules/gcs_utils.py` points to the GCS bucket `data_ecair_paaraal/raw/`. For local execution, pass a custom `file_map` to `load_all_assessments()` as shown above.

7.4 Adding New Timepoints

When new assessment data becomes available:

1. Place the raw CSV in `data/raw/`.

2. A

8. AI Disclosure

AI coding assistants (Claude, Anthropic) were used during the development of this project for code review, documentation drafting, and structured prompt generation. All methodological decisions, domain interpretations, and final outputs were made and validated by the ECAIR team. The underlying data, analytical logic, and policy framing reflect human judgment; AI tools served as a productivity aid, not a decision-maker.